

COLLEGE OF ENGINEERING, CHENGANNUR

Department of Computer Engineering

B.Tech Computer Science and Engineering | CSD415 Project Phase 1

PERIODIC PROJECT REPORT

AI-Powered Multimodal Misinformation Detection and Verification System

Project Group Number: 11

Team Members

- LCHN23CS203 — FATHIMA MARJAN M
- LCHN23CS207 — VAISHNAVI PRASAD
- CHN23CS147 — NEMUEL SAM JERRY
- CHN23CS192 — TWINKLE FRANCIS

Period 1 — 17 July to 30 July 2026: Topic Finalization, Approval Form, Reference Paper & Initial Planning

KTU Reg No.	Who?	Work Assigned	Work Done	Work Scheduled for Next 14 Days
—	Group	Finalize the project topic, understand the proposed system, begin literature review and complete initial project formalities.	Discussed the selected project topic with the guide and finalized the project direction. The team began literature review and project planning.	Complete initial literature review, collect the dataset and continue project documentation.
CHN23CS147	NEMUEL SAM JERRY	Contribute to topic selection, reference-paper study, documentation and project planning.	Participated in topic discussions, studied the reference paper and contributed to the initial literature review. Prepared the initial Project Approval Form/documentation.	Continue literature review and dataset collection; begin text-model implementation.
LCHN23CS207	VAISHNAVI PRASAD	Participate in topic selection, literature review and initial project setup.	Participated in project discussions and literature review and contributed to initial project planning.	Assist with dataset collection and model-development planning.
CHN23CS192	TWINKLE FRANCIS	Participate in project planning and literature review.	Participated in project discussions and initial literature review.	Begin planning the React.js frontend.
LCHN23CS203	FATHIMA MARJAN M	Participate in project planning and literature review.	Participated in project discussions and initial literature review.	Begin frontend planning and support project documentation.

Period 2 — 31 July to 13 August 2026: Dataset Collection, DistilBERT 10K & Frontend Development

KTU Reg No.	Who?	Work Assigned	Work Done	Work Scheduled for Next 14 Days
—	Group	Collect and preprocess the Fakeddit dataset, begin ML implementation and continue literature review.	The team worked on dataset collection and initial implementation of the text-based misinformation detection system.	Train the text model on a larger dataset, start image-model training and continue frontend development.
CHN23CS147	NEMUEL SAM JERRY	Download and prepare dataset and implement the initial text classifier.	Worked with Vaishnavi Prasad on downloading the Fakeddit dataset. Developed and trained the initial DistilBERT model using 10,000 samples.	Increase DistilBERT training to a larger dataset and continue ML implementation.
LCHN23CS207	VAISHNAVI PRASAD	Assist with dataset collection and begin image-model development.	Worked with Nemuel Sam Jerry on dataset downloading and preparation and began development of the EfficientNet-B0 image model.	Continue EfficientNet-B0 training.
CHN23CS192	TWINKLE FRANCIS	Begin frontend development.	Started development of the project webpage using React.js.	Continue React.js frontend development for the different input modalities.
LCHN23CS203	FATHIMA MARJAN M	Begin frontend development and assist with documentation.	Started development of the project webpage using React.js and contributed to project documentation.	Continue React.js frontend development and documentation.

Period 3 — 14 August to 27 August 2026: DistilBERT 50K, EfficientNet-B0 10K & Initial Progress Presentation

KTU Reg No.	Who?	Work Assigned	Work Done	Work Scheduled for Next 14 Days
—	Group	Improve the ML models, continue literature review and present project progress.	All four members worked together in the computer lab to train the DistilBERT model using 50,000 samples. The team also continued literature review and project discussions.	Continue image-model training, frontend development and preparation of literature-review materials.
CHN23CS147	NEMUEL SAM JERRY	Participate in large-scale text-model training and project presentation/documentation.	Worked with all team members in the computer lab on the DistilBERT 50K training. Conducted the initial project/literature-review presentation together with Vaishnavi Prasad, following the prepared literature-review PPT.	Continue image-model training, documentation and model integration work.
LCHN23CS207	VAISHNAVI PRASAD	Develop and train the image classification model and support presentation.	Developed and trained EfficientNet-B0 in Google Colab from scratch up to 10,000 samples. Conducted the presentation together with Nemuel Sam Jerry and contributed to literature review.	Support further image-model development and literature-review work.
CHN23CS192	TWINKLE FRANCIS	Continue frontend development and literature review.	Continued React.js frontend development and participated in	Continue frontend development and prepare for backend/model

			the literature review.	integration.
LCHN23CS203	FATHIMA MARJAN M	Continue frontend development and literature review.	Continued React.js webpage development and participated in the literature review.	Continue frontend development and support project documentation.

Period 4 — 28 August to 10 September 2026: EfficientNet-B0 20K, IEEE Literature Review & GitHub Work

KTU Reg No.	Who?	Work Assigned	Work Done	Work Scheduled for Next 14 Days
–	Group	Expand the image model, complete the literature review and organize project materials.	The team continued ML development and completed the literature review. Later, the literature review was converted into IEEE conference-paper format for publishing/documentation.	Proceed with multimodal fusion and continue frontend-backend integration.
CHN23CS147	NEMUEL SAM JERRY	Continue image-model training and handle repository/documentation work.	Trained EfficientNet-B0 on the additional 10,000 samples, extending the image model from Vaishnavi's initial 10K to approximately 20K total samples. Made later changes to the initially created GitHub repository, completed the GitHub documentation approval form, and worked on the reference paper and related documentation.	Begin multimodal feature fusion and continue project documentation.
LCHN23CS207	VAISHNAVI PRASAD	Support image-model development, literature review and repository setup.	Contributed to the literature review and initial GitHub repository creation and completed the first 10K EfficientNet-B0 training.	Assist with multimodal model integration and testing.
CHN23CS192	TWINKLE FRANCIS	Continue frontend development and literature review.	Continued development of the React.js frontend and contributed to the literature review.	Continue frontend refinement and prepare for backend integration.
LCHN23CS203	FATHIMA MARJAN M	Continue frontend development, literature review and documentation.	Continued development of the React.js webpage and contributed to the literature review and documentation.	Continue frontend-backend integration work and documentation.

Period 5 — 11 September to 14 September 2026: Current Implementation & Project Documentation

KTU Reg No.	Who?	Work Assigned	Work Done	Work Scheduled for Next 14 Days
–	Group	Consolidate completed work and proceed toward multimodal system integration.	The team has completed the major literature review, individual text and image model development and frontend development. Current project work is moving toward multimodal integration.	Implement feature-level multimodal fusion, develop PDF/video processing and integrate the frontend with the backend.
CHN23CS147	NEMUEL SAM JERRY	Continue ML integration and finalize current project documentation.	Continued implementation of the ML pipeline. Current models include the DistilBERT model trained on 50K samples and EfficientNet-B0 trained on approximately 20K image samples. Continued project documentation and repository work.	Implement the MLP-based multimodal feature-fusion component and continue backend integration.
LCHN23CS207	VAISHNAVI PRASAD	Support ML integration and project documentation.	Contributed to the completed literature review, image-model development and project documentation.	Assist with multimodal model integration and evaluation.
CHN23CS192	TWINKLE FRANCIS	Continue frontend development.	Continued development of the React.js frontend webpage for the project.	Connect the frontend with the FastAPI backend and support multimodal input interfaces.
LCHN23CS203	FATHIMA MARJAN M	Continue frontend development and project documentation.	Continued development of the React.js webpage and contributed to project documentation and literature review.	Continue frontend-backend integration and improve the user interface.